Roger Richard Demello

AI Engineer | Software Engineer | India | B.Tech 2026 | Immediate joiner
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Professional Summary

AI and software engineer with 3 internships building production **RAG** applications, **AI agents** with human approval, models measured against baselines, and full-stack services in **Python**, **TypeScript** and **Java**. 3 production applications, 200+ users, 99.5% uptime. AWS Certified, two-time AI hackathon placer.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++, C, Bash, YAML, Dart
Machine Learning: scikit-learn, PyTorch, LightGBM, Random Forest, IsolationForest, decision trees, anomaly detection
Model Evaluation: baselines, cross-validation, ROC-AUC, PR-AUC, NDCG@10, temporal splits, survey weighting, MLflow
LLMs and RAG: LLMs, RAG, prompt engineering, embeddings, semantic search, hybrid search, chunking, reranking
Vector Search: pgvector, FAISS, Pinecone, ChromaDB, ANN retrieval, two-tower embeddings, rank-bm25, in-memory search
Agents and Evals: AI agents, multi-agent systems, LangGraph, LangChain, LlamaIndex, MCP, human-in-the-loop, LLM evals
LLM Platforms: OpenAI, Azure OpenAI, Gemini, Claude, Ollama, Hugging Face, Sarvam AI, fine-tuning (conceptual)
Backend and APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, WebSockets, SSE, webhooks, Pydantic
Databases: PostgreSQL, SQL, SQLAlchemy, Prisma ORM, Redis, SQLite, MySQL, Supabase, Neo4j, AWS RDS
Frontend: React, Next.js, Preact, TypeScript, Tailwind CSS, Vite, shadcn/ui, Recharts, Streamlit, Flutter
Data and Notebooks: Pandas, NumPy, SciPy, statsmodels, matplotlib, Jupyter Notebooks, NetworkX, ETL, Celery, pg-boss
Cloud and DevOps: AWS, EC2, S3, IAM, CloudWatch, Cloud Run, Render, Vercel, Docker, Docker Compose, Linux
Delivery and Tools: CI/CD, GitHub Actions, Git, GitHub, Turborepo, Postman, Claude Code, Cursor, cloud-native design
Testing and Monitoring: pytest, Vitest, JUnit 5, Mockito, Playwright, Prometheus, Grafana, Sentry, telemetry, audit logging
Security and Access: OAuth, OIDC, JWT, RBAC, field-level encryption, HMAC webhook signatures, bcrypt, consent tracking
Java and Concurrency: Java 17, ExecutorService, ReentrantLock, JDBC, HikariCP, Maven, design patterns, multithreading
Embedded and Edge: ESP32, ESP32-CAM, Arduino, RS485, Modbus RTU, MAX485, UART, I2C, SPI, TensorFlow Lite
Integrations: Gmail API, Slack API, HubSpot API, Salesforce API, WhatsApp Cloud API, Instagram Graph API
Fundamentals: data structures, algorithms, object-oriented design, system design, incident response, root cause analysis

Work Experience

Innovun Global August 2026 – Present
AI Engineer Intern
- Build the Auri chat widget in **Preact** for Cruz Roja Mexicana (Red Cross, Mexico), a bilingual **RAG** assistant for 67 courses
- Implement **hybrid search** in the **RAG** pipeline, fusing **pgvector** vector and keyword scores with reciprocal rank fusion
- Develop a PO-to-SO proof of concept that extracts PDF fields with **OCR** and **Gemini**, gated by **human-in-the-loop** review
- Test extraction on **13 sample PO layouts**, including scans with no text layer, skewed pages and multi-column pages

AI LifeBOT January 2026 – June 2026
AI Engineer Intern
- Architected **RAG** pipelines in **3 production applications** serving **200+ users** on **Python**, FastAPI and PostgreSQL
- Cut query and response **latency 35%** by profiling hot paths across the request path of an inherited codebase
- Sustained **99.5% uptime** with telemetry, structured **audit logging**, incident response and root cause analysis
- Automated **pytest** suites in **CI/CD** pipelines, shipping 5+ features end to end through cross-functional collaboration

CFM, RCOEM May 2025 – July 2025
Machine Learning Research Intern
- Engineered features in **Python** and **Pandas** from **1,000+** raw health records, cleaning and validating data
- Trained a **Scikit-learn** clinical **classification** model reaching **87% accuracy** on held-out records
- Raised validation reliability **25%** using **cross-validation** across multiple splits, not a single sample
- Standardized **data pipelines** into reusable, verified components, cutting development time **30%**

Projects

Autonomous Executive Email Copilot | *Python, FastAPI, SQLAlchemy, PostgreSQL, Azure OpenAI, Docker* | Live | GitHub
- Engineered an inbox copilot that triages Gmail and Microsoft 365 over **OAuth**, infers priority, deadline and risk, and holds every outbound action for **human approval**
- Designed the **LLM** as prose-only with deterministic routing and fact-checked drafts, gated by a **CI rubric** that caught an invented deadline
- Benchmarked 3 **agents** on a Gym-style environment over **27 episodes** with bounded graders, publishing weak results at about $0.009 per LLM episode
- Secured the multi-tenant API with **OIDC** single sign-on, 3 ranked roles, encrypted mailbox tokens, a per-org **audit log**, data export and hard delete

ML Guardian | *Python, FastAPI, MCP, DataHub, Google Gemini, pytest, GitHub Actions* | GitHub
- Built an **agent** on DataHub that detects **3 failure types** (stale upstreams, null-rate creep, renamed columns) before a KPI moves
- Implemented lineage tracing that names the downstream models and dashboards at risk, then generates fail-fast remediation code
- Integrated mcp-server-datahub over the Model Context Protocol (**MCP**) and packaged the workflow as a reusable DataHub Skill
- Implemented an **offline-first** mode on bundled fixtures, validated by 9 **pytest** tests and **GitHub Actions** CI

music-recsys | *Python, PyTorch, LightGBM, FastAPI, MLflow, Redis, Prometheus, Grafana, Docker* | GitHub
- Developed a 2-stage recommender: **two-tower embeddings** feed **ANN retrieval** into a **LightGBM** ranker served by FastAPI
- Lifted **Recall@10** from 0.094 to 0.105 and **NDCG@10** from 0.084 to 0.094 on a held-out temporal split of synthetic data
- Tracked every run in **MLflow** and checked **HNSW** ANN recall@20 of 1.0 against the exact index
- Decoupled **5 components** (cache, event bus, store, ANN index, registry) behind **Protocol** interfaces with swappable backends
- Designed retrain and embedding-refresh jobs and an A/B simulator, monitored with **Prometheus** and **Grafana**

LifePulse | *Python, Flask, scikit-learn, pandas, NumPy, SciPy, joblib* | Live | GitHub
- Trained **HistGradientBoosting** heart-disease and migraine models, reaching **ROC-AUC 0.855** on **312k** US adults and 84.0% accuracy against a 60.0% baseline
- Replaced a RandomForest model reporting **R-squared 0.81** after finding its training data was generated, not observed
- Applied **survey weights** that corrected a **25%** overstatement of heart-disease prevalence (0.090 unweighted vs 0.072)
- Measured an emergency check on **110** labelled phrasings, cutting false alarms from **46.2%** to **10.8%** at 100% sensitivity

sentinelops | *Python, FastAPI, scikit-learn, statsmodels, NetworkX, React, TypeScript* | GitHub
- Built **IsolationForest anomaly detection** over synthetic telemetry to predict incidents before they occur
- Orchestrated **multi-agent** root-cause analysis with human-approved remediation and **audit logging**
- Mapped the blast radius of each failure with a **NetworkX** dependency graph and estimated business impact
- Raised Slack and PagerDuty alerts and simulated self-healing behind human approval

contentflow-ai | *Python, LangGraph, scikit-learn, FastAPI, Streamlit, Azure OpenAI, Redis, pytest, Docker* | GitHub
- Orchestrated a **multi-agent** content engine in **LangGraph** covering drafting, compliance checking, localization and publishing
- Validated content against policy documents with local **RAG**, then remediated automatically and recorded an **audit log**
- Added **human-in-the-loop** approval gates, Hindi localization with offline fallbacks and an ROI model estimating about Rs 2.1 Cr per year

DealSentry | *TypeScript, React, Express.js, Prisma, PostgreSQL, Azure OpenAI, Vitest* | Live | GitHub
- Built a proposal compliance and risk-review system with a rules engine, risk scoring and **role-based access control**
- Integrated Salesforce, HubSpot and Gmail over **OAuth 2.0** and used **Azure OpenAI** for analysis, with discount and payment limits re-checked in code
- Implemented embedding-based **semantic search** on **pgvector** with a substring fallback and PDF export, tested with Vitest and Supertest

MARK - AI HR Automation Agent | *Python, FastAPI, PostgreSQL, Redis, Celery, React, TypeScript* | GitHub
- Built an HR platform pairing an employee chat widget with an HR analytics dashboard and **multi-agent** orchestration for sentiment, nudges and complaints
- Grounded policy Q&A with **hybrid search** that combines rank-bm25 and vector similarity over documents read with pypdf
- Added **field-level encryption**, audit logging and WhatsApp and calendar integrations

Shadow GTM | *Next.js 16, TypeScript, React 19, Supabase, Gemini API, Zod, Vitest* | GitHub
- Built a competitor-intelligence platform that diffs each scan against the last and extracts signals with a verbatim source quote
- Designed a multi-tenant **Supabase** schema with organizations, roles and **row-level security** on every table
- Streamed Gemini battlecards over **SSE** with structured outputs validated by Zod, with a mock mode on built-in sample pages

navi-paytm (Paytm Merchant Copilot) | *Flutter, FastAPI, SQLAlchemy, SQLite, Sarvam AI, Telegram Bot API* | GitHub
- Built with one other engineer a mobile assistant that turns photos, voice notes and chat into product catalogs, orders and stock alerts
- Delivered multilingual AI across 23 Indian languages using Sarvam OCR, speech-to-text, LLM and text-to-speech APIs
- Shipped 10+ modules including Auto-Catalog, Telegram Order Inbox, Stock-Out Predictor, Banner Generator and Copilot Chat

FinCore - Concurrent Banking Engine | *Java 17, JDBC, PostgreSQL, HikariCP, JUnit 5, Mockito, Docker* | GitHub
- Engineered a **Java 17** banking backend with multi-currency transfers, scheduled payments and investment portfolios
- Built a concurrent engine with **ExecutorService** pools and ID-ordered **ReentrantLocks**, stress-tested under 20+ threads
- Guaranteed **ACID** JDBC transactions with row locking, client-token idempotency and **HikariCP** pooling on PostgreSQL
- Designed pluggable fraud detection with the Strategy pattern, plus RBAC, bcrypt hashing and Observer-driven audit logging
- Passed 56 tests in **JUnit 5**, **Mockito** and H2 on CI, with TCP socket and REST interfaces and Docker delivery

NPK Crop Recommendation System | *Python, micromlgen, TensorFlow Lite, Arduino C++, ESP32, RS485* | GitHub
- Integrated an **RS485** NPK soil sensor, a **MAX485** converter and an **ESP32** into one working **Modbus RTU** chain
- Documented the **wiring diagram** and RS485 troubleshooting so the hardware could be rebuilt and debugged
- Trained a **Decision Tree** in Python and deployed it to the ESP32 as a C++ header, plus **int8 TensorFlow Lite** variants
- Ran inference **fully on-device** with no internet, so the recommendation happens in the field

Automatic Solar Panel Cleaning System | *FastAPI, OpenCV, Arduino C++, ESP32, ESP32-CAM, TB6612* | GitHub
- Integrated 2 microcontrollers and a server into one automated monitor-and-clean cycle over **WiFi**
- Captured a panel image every 60 seconds with an **ESP32-CAM** and detected dust with **OpenCV** thresholds
- Drove the cleaning mechanism through a **TB6612** motor driver, with 2 **IR sensors** for edge detection
- Started and stopped the cleaner over **HTTP** commands served by the ESP32, debugging with serial logs

ImageToToonArt | *Python, FastAPI, OpenCV, PyTorch, React* | GitHub
- Built an image app that turns photos into cartoon artwork in 10 styles using **OpenCV** pipelines and optional **PyTorch** models
- Delivered a drag-and-drop **React** UI with before and after comparison and downloads through a **FastAPI** REST API

LabLingo AI | *Python, Flask, Ollama (llama3.2), React, Vite, Tailwind CSS* | GitHub
- Built a scientific dictionary that runs fully offline on a local **LLM** through Ollama, with no API keys
- Added user authentication and cached repeat queries behind a Flask API with a React and Tailwind UI

Echoes | *Next.js 14, TypeScript, Murf API, Gemini API, React Three Fiber* | Live | GitHub
- Built a Next.js app that turns personal memories into narrated stories in expressive Murf AI voices, with 7 story styles
- Added a dual-voice mode, emotion-tuned ambient music and multilingual dubbing in 5 languages

Portfolio | *Next.js 16, TypeScript, Tailwind CSS, Vercel* | Live | GitHub
- Built a responsive dark-theme portfolio with a particle background, 3D card tilt and SEO and Open Graph support, deployed on Vercel

Education

Shri Ramdeobaba College of Engineering and Management (RCOEM) 2022 – 2026
B.Tech, Electronics & Communication Engineering **GPA: 8.90/10**
Minor: **Artificial Intelligence & Machine Learning** (2022 – 2026) **GPA: 9.67/10**

Certifications and Awards

AWS Certified Cloud Practitioner – Amazon Web Services (October 2025)
Hackathon Finalist – Paytm x Sarvam x Logitech AI National Hackathon
Second Place – ByteSize Sage AI National Hackathon